A. A.M. Deviation

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: math, number theory | https://codeforces.com/problemset/problem/1605/A

A number a_2 is said to be the arithmetic mean of two numbers a_1 and a_3, if the following condition holds: $a_1 + a_3 = 2 \cdot a_2$.

We define an arithmetic mean deviation of three numbers a_1, a_2 and a_3 as follows: $d(a_1, a_2, a_3) = |a_1 + a_3 - 2 \cdot a_2|$.

Arithmetic means a lot to Jeevan. He has three numbers a_1, a_2 and a_3 and he wants to minimize the arithmetic mean deviation $d(a_1, a_2, a_3)$. To do so, he can perform the following operation any number of times (possibly zero):

- Choose i, j from $\{1, 2, 3\}$ such that $i \neq j$ and increment a_i by 1 and decrement a_j by 1

Help Jeevan find out the minimum value of $d(a_1, a_2, a_3)$ that can be obtained after applying the operation any number of times.

Input

The first line contains a single integer t ($1 \leq t \leq 5000$) — the number of test cases.

The first and only line of each test case contains three integers a_1, a_2 and a_3 ($1 \leq a_1, a_2, a_3 \leq 10^8$).

Output

For each test case, output the minimum value of $d(a_1, a_2, a_3)$ that can be obtained after applying the operation any number of times.

Example

Input
3 3 4 5 2 2 6 1 6 5

Output
0 1 0

Note

Note that after applying a few operations, the values of a_1, a_2 and a_3 may become negative.

In the first test case, 4 is already the Arithmetic Mean of 3 and 5.

$d(3, 4, 5) = |3 + 5 - 2 \cdot 4| = 0$

In the second test case, we can apply the following operation:

(2, 2, 6) xrightarrow[incrementa_2]decrementa_1 (1, 3, 6)

$d(1, 3, 6) = |1 + 6 - 2 \cdot 3| = 1$

It can be proven that answer can not be improved any further.

In the third test case, we can apply the following operations:

(1, 6, 5) xrightarrow[incrementa_3]decrementa_2 (1, 5, 6)
xrightarrow[incrementa_1]decrementa_2 (2, 4, 6)

$d(2, 4, 6) = |2 + 6 - 2 \cdot 4| = 0$